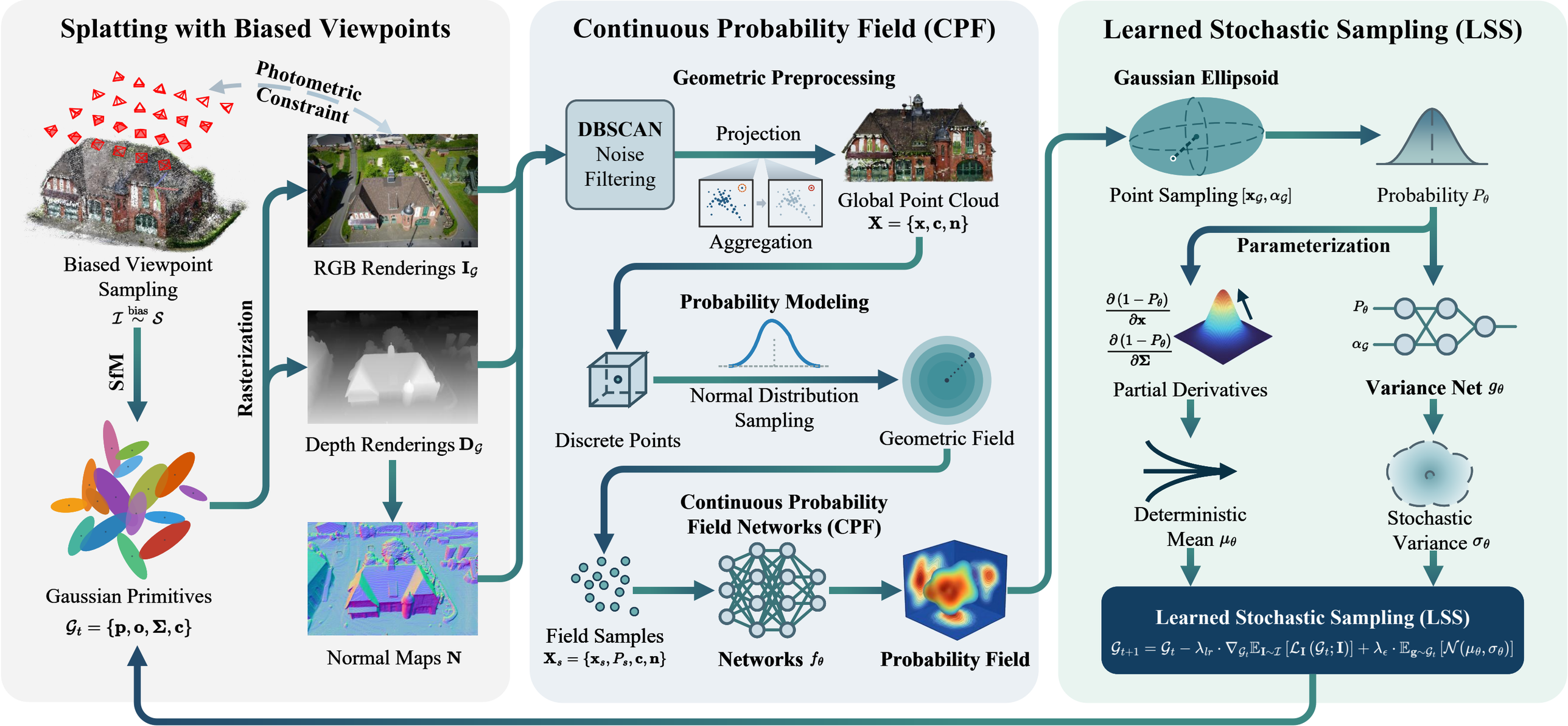

Splatting with Biased Viewpoints
Photometric Constraint
Biased Viewpoint Sampling
$\mathcal{I} \overset{\text{bias}}{\sim} \mathcal{S}$
SfM
Rasterization
RGB Renderings $\mathbf{I}_{\mathcal{G}}$
Depth Renderings $\mathbf{D}_{\mathcal{G}}$
Normal Maps $\mathbf{N}$
Gaussian Primitives
$\mathcal{G}_t = \{\mathbf{p}, \mathbf{o}, \mathbf{\Sigma}, \mathbf{c}\}$
Continuous Probability Field (CPF)
Geometric Preprocessing
DBSCAN Noise Filtering
Projection
Aggregation
Global Point Cloud
$\mathbf{X} = \{\mathbf{x}, \mathbf{c}, \mathbf{n}\}$
Probability Modeling
Discrete Points
Normal Distribution Sampling
Geometric Field
Continuous Probability Field Networks (CPF)
Field Samples
$\mathbf{X}_s = \{\mathbf{x}_s, P_s, \mathbf{c}, \mathbf{n}\}$
Networks f_θ
Probability Field
Learned Stochastic Sampling (LSS)
Gaussian Ellipsoid
Point Sampling $[\mathbf{x}_{\mathcal{G}}, \alpha_{\mathcal{G}}]$
Probability P_θ
Parameterization
$\frac{\partial(1-P_\theta)}{\partial \mathbf{x}}$
$\frac{\partial(1-P_\theta)}{\partial \mathbf{\Sigma}}$
Partial Derivatives
P_θ
$\alpha_{\mathcal{G}}$
Variance Net g_θ
Deterministic Mean μ_θ
Stochastic Variance σ_θ
Learned Stochastic Sampling (LSS)
$\mathcal{G}_{t+1} = \mathcal{G}_t - \lambda_{lr} \cdot \nabla_{\mathcal{G}_t} \mathbb{E}_{\mathbf{I} \sim \mathcal{I}} [\mathcal{L}_{\mathbf{I}}(\mathcal{G}_t; \mathbf{I})] + \lambda_\epsilon \cdot \mathbb{E}_{\mathbf{g} \sim \mathcal{G}_t} [\mathcal{N}(\mu_\theta, \sigma_\theta)]$